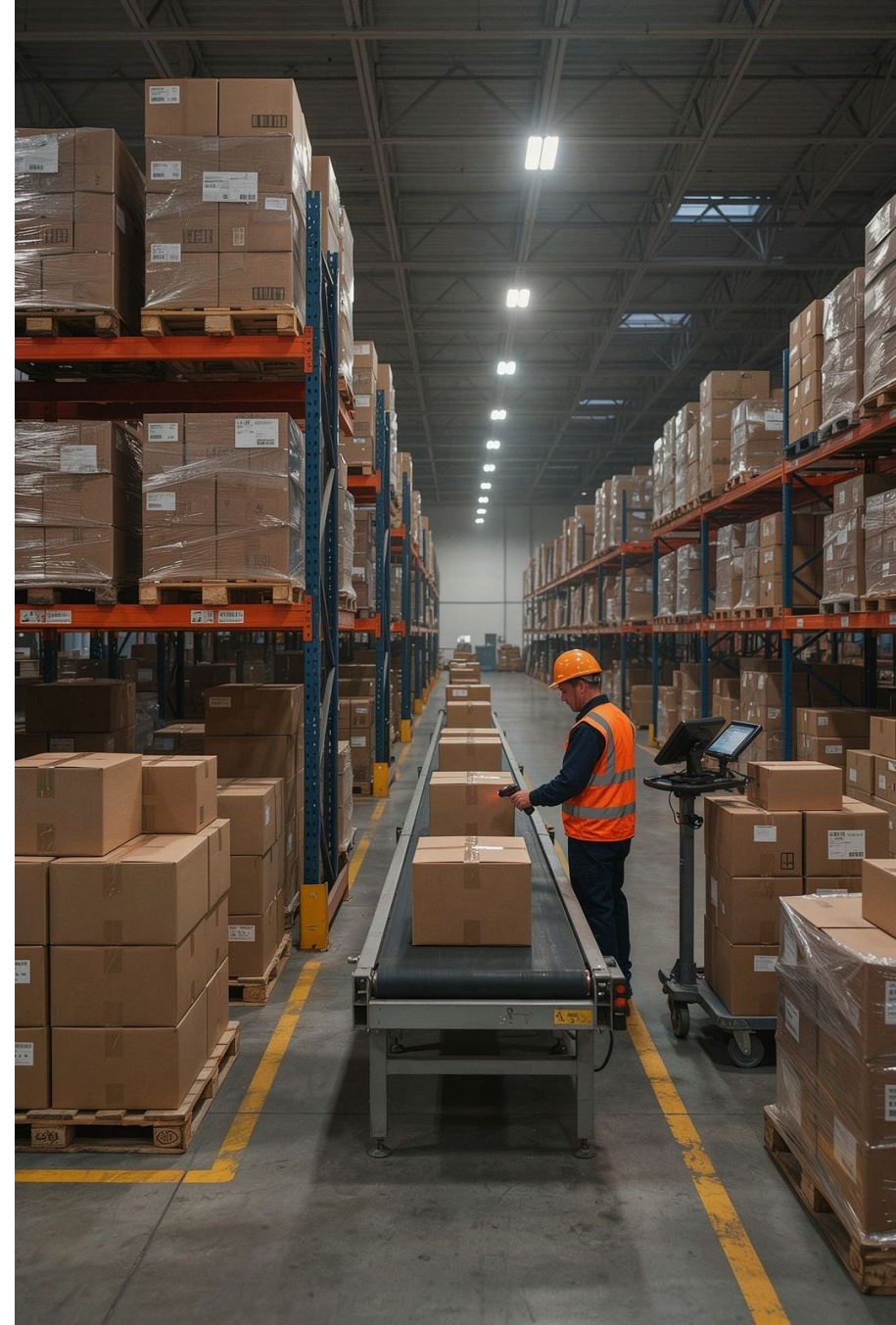


Outlier Analysis: The Interquartile Range Method

Finding the values that do not belong — deciding what they are, and deciding what to do about them.

Case study: **Meridian Retail** — 5,000 orders across three fulfilment hubs. A COO convinced that all hubs perform the same. Data that tells a very different story.



Why This Matters

Every dataset contains values far from the rest. What you do with them changes your answer — and the cost of getting it wrong runs in both directions.

Delete when you should not

- A bank removes its largest transactions as "errors" — and deletes the fraud it was hunting
- An insurer discards catastrophic claims and underprices every policy it writes
- A hospital drops its sickest patients and reports excellent outcomes

Keep when you should not

- One value mistyped as 9,900 instead of 99 moves the average order value by 50%
- A sensor stuck at 999 corrupts every summary statistic built on that column

 Outlier handling is a decision that needs a reason. It is not a cleanup step.

An Outlier Is Not the Same as a Mistake

Three distinct kinds of outlier exist in practice — and each one demands a different response. Most analysts treat all three as the first kind. That is where the damage happens.

1

A Data Error

A typo, a unit mix-up, a sensor fault, a placeholder like 999 or -1. The value does not represent anything real.

→ **Fix it, or remove it and document why.**

2

Rare but Genuine

A real observation from the same population that happens to be extreme. A genuinely large order. A genuinely slow delivery. The data is correct.

→ **Keep it. It is telling you about your tail risk.**

3

From a Different Population

The value is perfectly normal for a group you did not realise was mixed into your data. The outlier is a symptom of an unexamined segment, not an error.

→ **Segment the data. Delete nothing.**

In practice, most analysts assume they are dealing with type 1. In practice, most flags turn out to be type 2 or type 3. This deck ends with a live example of type 3 hidden inside Meridian's order data.

Three Questions to Ask About Any Flagged Value

Answer these in order, before touching anything. The questions form a decision tree. Only after you have answered all three should you decide what to do with the value.

1 Is it possible?

A delivery time of -3 days is impossible.
A weight of 0 kg is impossible. An order quantity of -50 units is impossible.
Impossible means error — no further investigation required before acting.

2 Is it plausible for this business?

A 45-day delivery is possible in principle.
Does Meridian have a process — international shipping, a bespoke item, a long back-order — that would produce one? If yes, it is real data from a real event.

3 Would removing it change my conclusion?

If the analysis result is identical with and without this value, the decision matters less. If removing it flips the finding, it matters enormously — and the decision **must be written down** and shared with any reader of the output.

 The default is keep. Removing data requires justification. Keeping it does not.

Quartiles: Cutting the Data into Four

Sort the data from smallest to largest. Find the three values that split the sorted list into four groups of equal size. These three split-points are the quartiles.

Q1 — 25th Percentile

25% of the data falls below this point. One quarter of observations sit to the left of Q1.

Q2 — 50th Percentile

50% falls below. This is the **median** — the middle value of the sorted dataset.

Q3 — 75th Percentile

75% falls below. Three quarters of observations sit to the left of Q3.

These are **positions**, not arithmetic on the values. Q3 does not care how large the largest value is — only that it sits above Q3. That property is the entire reason this method survives extreme values. **The measuring instrument cannot be bent by the thing it is measuring.** Standard deviation does not share this property, which is why it fails on skewed data.

Finding Quartiles by Hand

Meridian order values, **n = 9** (₹ thousands), sorted: 12, 14, 15, 15, 18, 20, 22, 25, 99

Position Formulae

Q1 position = $0.25 \times (n - 1) + 1$

Q3 position = $0.75 \times (n - 1) + 1$

When a position lands between two values, interpolate proportionally between them. All statistical software does this automatically — you only need the formula to understand what the software is reporting.

Calculation for n = 9

Quartile	Position	Result
Q1	$0.25 \times 8 + 1 = 3\text{rd value}$	15
Q2 (Median)	5th value	18
Q3	$0.75 \times 8 + 1 = 7\text{th value}$	22

Notice that the value 99 plays no role whatsoever in determining Q1, Q2, or Q3. Its presence or absence has zero effect on these positions.

The Interquartile Range

Definition

$$\text{IQR} = Q3 - Q1$$

The width of the middle 50% of the data. It is a measure of spread that is completely immune to extreme values in either tail.

For our nine orders: **IQR = 22 – 15 = 7**

The Key Property

Change the value 99 to 9,900 — a hundredfold increase — and the IQR is still exactly 7. Q1 and Q3 do not move. Nothing about the middle 50% of the sorted data changes when you inflate a value that was already outside it.

Comparison with Standard Deviation

On the same nine values, the standard deviation is **$s = 27.44$** — larger than the median itself of 18. The single value of 99 inflated it so severely that σ is no longer describing typical variation; it is describing the one extreme case.

The IQR reports the true spread of the bulk of the data. The standard deviation has been hijacked by the very value it was meant to help detect.

- This is why the IQR is the right foundation for outlier detection on real-world business data, which is almost never perfectly symmetric.

The Fence Formula

Once you have the IQR, calculating the detection boundaries is a single arithmetic step on each side.

Lower Fence

$$Q1 - 1.5 \times IQR$$

For our nine orders:

$$15 - (1.5 \times 7) = 15 - 10.5 = \mathbf{4.5}$$

Upper Fence

$$Q3 + 1.5 \times IQR$$

For our nine orders:

$$22 + (1.5 \times 7) = 22 + 10.5 = \mathbf{32.5}$$

Anything outside the fences is **flagged**. In our nine-value dataset, 99 exceeds the upper fence of 32.5. Nothing falls below 4.5.

⊗ Flagged means "go and look at this." It does not mean delete. The fence is a referral mechanism, not an execution order.

Where 1.5 Comes From

The multiplier 1.5 is not arbitrary, but it is also not a law of nature. It is a deliberate calibration decision grounded in the behaviour of normal data.

The Statistical Reasoning

If data follows a normal distribution, the IQR equals approximately **1.349 standard deviations**. Multiplying 1.5 by 1.349 gives 2.024, which added and subtracted from both sides places the fences at **± 2.70 standard deviations** from the mean.

At $\pm 2.70\sigma$ on a normal distribution, the proportion of data outside the fences is **0.70%** — roughly 1 observation in every 143. This is frequent enough to catch genuine problems, rare enough not to generate an overwhelming number of false alarms.

The 3.0 Alternative

Using a multiplier of 3.0 instead of 1.5 moves the fences to **± 4.72 standard deviations**, flagging approximately 1 observation in every 427,000 on normal data. Essentially nothing gets flagged unless it is a catastrophic deviation.

Multiplier	σ equivalent	Flag rate	Convention
1.5	$\pm 2.70\sigma$	0.70%	Outlier
3.0	$\pm 4.72\sigma$	~0.0002%	Extreme outlier

If a column flags far more than about 1% of its values, the data is not well-behaved — check the assumption before blaming the data.

Worked Example: Full Calculation

Data: 12, 14, 15, 15, 18, 20, 22, 25, 99

Step	Calculation	Result
Q1	3rd value in sorted list	15
Q3	7th value in sorted list	22
IQR	$22 - 15$	7
$1.5 \times \text{IQR}$	1.5×7	10.5
Lower fence	$15 - 10.5$	4.5
Upper fence	$22 + 10.5$	32.5
Flagged	Values outside [4.5, 32.5]	99

 Note what made this work: the value 99 played no part in setting the fence that caught it. Q1 and Q3 were fixed by position alone, so the outlier could not raise the bar it then had to clear. This is the core advantage of the IQR method over z-scores.

Your Turn — Work This Out

A support desk logs ticket resolution times, in hours. Thirteen tickets, already sorted:

2, 3, 3, 4, 4, 5, 5, 6, 6, 7, 8, 9, 31

Compute the following in order. Five minutes.

01

Q1

Position = $0.25 \times (13 - 1) + 1$

02

Median

Position = middle value of 13

03

Q3

Position = $0.75 \times (13 - 1) + 1$

04

IQR, Both Fences

Apply the standard formula with $1.5 \times \text{IQR}$

05

Flag & Whiskers

Which values are flagged? Where does each whisker end?

Pay particular attention to what happens with the lower fence. If it produces a surprising result, that is deliberate — it is the point of the exercise.

Exercise Answers

Data: 2, 3, 3, 4, 4, 5, 5, 6, 6, 7, 8, 9, 31

Step	Working	Answer
Q1	4th value	4
Median	7th value	5
Q3	10th value	7
IQR	$7 - 4$	3
Lower fence	$4 - 4.5$	-0.5
Upper fence	$7 + 4.5$	11.5
Flagged above 11.5	$31 > 11.5$	31
Whiskers	Furthest real values inside fences	2 and 9

On the Negative Lower Fence

The lower fence of -0.5 is correct, not an error. A resolution time cannot be below zero, so nothing can ever be flagged at the low end. That is the formula telling you something true: on this data, short resolution times are never statistically unusual. The fence reached into an impossible region and found nothing there — which is exactly the right result.

On the Upper Whisker

The upper whisker stops at **9**, not at 11.5. The whisker always ends at the largest *actual value* inside the fence, never at the fence itself. There were no observations between 9 and 11.5, so the whisker terminates at 9. The gap between 9 and 31 is where the outlier lives, visually.

The Complete Six-Step Recipe

Works on any numeric column in any dataset. Steps 1–5 are arithmetic; Step 6 is the analysis — and it is the step that gets skipped.

1

Sort

Order the data from smallest to largest

2

Find Q1 and Q3

Use position formulae or `.quantile()`

3

Compute IQR

$IQR = Q3 - Q1$

4

Set Fences

$Lower = Q1 - 1.5 \times IQR$ · $Upper = Q3 + 1.5 \times IQR$

5

Flag

Everything outside [Lower, Upper]

6

Investigate Every Flag

Error, rare-but-real, or different population?

⊗ A pipeline that runs steps 1–5 and auto-deletes is not doing outlier analysis. It is destroying data on a schedule. Flag and quarantine. Never auto-delete.

Anatomy of a Box Plot

A box plot is the IQR method drawn. Every number from the six-step recipe appears in it. Reading one fluently takes less than a minute once you know what each element encodes.

Elements and What They Represent

The Box

Spans Q1 to Q3 — the middle 50% of the data. The width of the box is the IQR, readable at a glance without any calculation.

The Centre Line

The median (Q2). Its position inside the box tells you about skewness: centred means symmetric; pushed towards one end means the tail stretches the other way.

The Whiskers

Extend from the box edges to the **furthest actual data points still inside the fences**. A short whisker means there was no data out there — not that the fence is unusually tight.

The Dots

Any observation outside a fence is plotted individually as a point. These are the flagged values — not confirmed errors, but candidates for investigation.

Critical Distinction

- ❏ The whisker is not the fence. In our worked example, the fence is at 32.5 but the whisker stops at 25, because 25 is the largest real value inside the fence. When you read a box plot, the whisker tip is always a data point — the fence is invisible.

For our nine-value dataset:

- Box: 15 to 22 (width = IQR of 7)
- Median line at 18
- Upper whisker ends at 25
- Single dot at 99 — the flagged value

Reading a Box Plot

Four analytical questions answered instantly from a single glance, with no arithmetic required from the reader.



Centre

Where does the median line sit? This is your robust measure of the typical value, unaffected by any extreme values in either tail.



Spread

How wide is the box? A narrow box means the middle 50% is tightly clustered. A wide box means substantial variability across the bulk of the data.



Skewness

Where does the median sit *inside* the box? Centred → symmetric. Pushed towards Q1 → right-skewed with a long upper tail. Pushed towards Q3 → left-skewed.



Outliers

How many dots appear, and how far do they reach? The distance of a dot from the fence tells you how extreme the value is relative to the rest of the distribution.

One picture answers three of the four questions from a typical descriptive statistics session — centre, spread, and skewness simultaneously — while also surfacing every flagged value. A table of summary statistics cannot show you skewness or the distribution of extreme values at a glance.

Where Box Plots Earn Their Place

A single box plot is adequate for understanding one column. **Side-by-side box plots are excellent** — this is where the format genuinely outperforms every alternative.

Ideal Comparison Scenarios

- Delivery times across fulfilment hubs
- Salary distributions across departments or seniority bands
- Response time before and after a process change
- Order value across product categories or customer segments
- Resolution time across support agents or shifts

You get centre, spread, skew, and outliers for every group simultaneously, on one shared scale. The comparison is visual and immediate.

When to Use a Table Instead

Two or three groups and one metric → a table of summary statistics is faster and sufficient. The reader does not need a chart to compare three numbers.

More than three groups, or when you need to see distributional shape → use box plots. The visual encoding of spread and skewness becomes essential as the number of groups grows.



Rule of thumb: two to three groups and one metric, use a table. More than that, use box plots.

CASE STUDY

The Case: Meridian's Three Hubs

Meridian ships from three fulfilment hubs — Mumbai West, Bengaluru South, and Delhi NCR. The COO wants to know which hub has a reliability problem. She has already been told by a colleague, based on mean delivery times, that all three perform the same.

The Data

- 5,000 orders with `delivery_days` recorded per order
- Each order assigned to exactly one hub
- No missing values in the delivery column

The Complication

All three hubs average between **3.28 and 3.39 days**. On the mean alone they are indistinguishable, separated by a total of 0.11 days across three hubs serving 5,000 orders.

The mean has given the COO a false picture of uniformity. Outlier analysis and the IQR will tell a very different story.

Applying the Method to Each Hub

Run the six-step recipe independently on each hub's delivery times. The results reveal structure that the mean completely concealed.

Metric	Mumbai West	Bengaluru South	Delhi NCR
Orders	2,017	1,717	1,266
Q1	3.00	2.00	1.20
Median	3.30	3.10	2.70
Q3	3.60	4.20	4.10
IQR	0.60	2.20	2.90
Upper fence	4.50	7.50	8.45
Flagged	44 (2.18%)	21 (1.22%)	49 (3.87%)
Worst delivery	24.2 days	54.6 days	44.3 days

❏ Read the IQR row first. Mumbai's middle 50% spans 0.6 days. Delhi's spans 2.9 — nearly five times wider. Same average delivery time. Completely different reliability profile.

The Result That Changes How You Read Outlier Counts

Mumbai West is the best-performing hub — and it flags more than twice as many outliers as Bengaluru South. This apparent contradiction is the most important result in the entire analysis.

Why the Counts Are Misleading

Each hub's fence is built from its own IQR:

- Mumbai's fence sits at **4.50 days**, because its IQR is only 0.60
- Delhi's fence sits at **8.45 days**, because its IQR is 2.90

A 5-day delivery is an outlier in Mumbai and completely ordinary in Delhi. Mumbai's tight process sets a tight fence, so ordinary variation trips it. Delhi's wide fence lets serious delays pass unnoticed.

The Correct Interpretation

Mumbai West has the highest raw outlier count *because* it is the most consistent hub. Its fence is placed so close to the bulk of its deliveries that even moderate deviations cross it.

Delhi NCR's lower count of flagged orders does not mean it performs better — it means its baseline is already so variable that severe delays are not statistically unusual there.

⊗ Never compare raw outlier counts across groups. Always compare the IQRs and the fences that produced them. The count alone is meaningless without context.

Investigating the Flags

Company-wide, 205 orders were flagged at a consolidated fence of 5.9 days. Step 6 of the recipe: what are these orders? Are they errors, genuine extremes, or a different population?

Where They Came From

Hub	Flagged Orders
Delhi NCR	121
Bengaluru South	79
Mumbai West	5

What Happened to Those Customers

Metric	Flagged Orders	Everything Else
Average rating	3.30	3.67
Return rate	15.1%	14.6%

Verdict: rare but genuine. Not data errors. These are real deliveries that went badly, concentrated heavily in Delhi NCR and Bengaluru South, costing Meridian a third of a star in customer satisfaction rating. There is nothing to fix, correct, or remove in the dataset. The flagged orders are the finding.

❌ Deleting these 205 orders would have removed the exact evidence the COO asked for. The dataset would have looked cleaner. The conclusion would have been completely wrong.

The Decision

Ranking by average delivery time separates the three hubs by at most 0.11 days — a difference that is operationally meaningless and statistically trivial. The mean told the COO that nothing was wrong. The IQR tells a different story entirely.

Route Time-Critical Orders Through Mumbai West

Not because it is faster on average — all three hubs are essentially identical on that metric. But because it is **predictable**. A delivery promise is a statement about the worst case, not the average. Mumbai's IQR of 0.60 days makes its worst-case performance foreseeable.

Investigate Delhi NCR Operationally

It produces 3.87% severe delays against Mumbai's 5 orders out of 2,017. The wide IQR of 2.90 days indicates a process that is fundamentally inconsistent, not merely unlucky on a few orders.

Keep Every Flagged Order in the Dataset

The 205 flagged orders are the finding, not noise obscuring it. They carry the customer satisfaction signal, the hub-level comparison, and the operational evidence for the investigation recommendation.

You Have Flagged Them. Now What?

Six options are available once a value is flagged. Deleting is one of six — not the default, not the first resort, and often not the right choice at all.

1. Keep Them

The default. Report the median and IQR alongside the mean so the tail is visible in your output without removing any observations. Use descriptive language to characterise the distribution's shape.

2. Fix Them

Only when you can establish the true value with confidence. A confirmed unit error, a known sensor fault, a documented typo where the original source is available. Fix requires evidence, not inference.

3. Cap Them (Winsorise)

Replace anything above the upper fence with the fence value itself. Keeps the row and all its other columns in the dataset; limits the distortion on aggregate statistics without deleting any record.

4. Transform the Column

A log transform pulls in a long right tail across the entire column. On the Meridian order value column it reduced skewness from 12.2 to 0.48 in one operation — making standard parametric methods applicable.

5. Segment

If the flagged values form a coherent group — corporate customers, a specific product category, a particular hub — split the data and analyse each group separately. The outliers dissolve; the structure becomes visible.

6. Remove Them

Last resort, and only for confirmed errors. Always documented. Always reported to any reader of the analysis. The number removed and the basis for removal must appear in the methodology section.

Choosing a Treatment

Match the treatment to what the flag turned out to be. The diagnostic question comes first; the treatment follows from the answer.

What the flag turned out to be	Treatment
Impossible value (negative time, weight of zero, placeholder 999 or -1)	Fix or remove — and document both the decision and the count
Genuine extreme, and it <i>is</i> the subject of the analysis	Keep. It is your finding. Removing it deletes the study.
Genuine extreme, distorting a summary statistic	Report median and IQR beside the mean. Let the reader see both
Genuine extreme, breaking a downstream model	Cap or transform — do not delete the row
A whole group of them, belonging together	Segment the data and analyse each group separately
Cannot tell — insufficient information to classify	Keep it and say so. Uncertainty is a reportable finding.

 Whatever you choose, state it. Any analysis that excluded or altered observations must report how many and on what basis. Without that, nobody can verify the result — including you, six months from now.

What the Combined View Hides

Run the IQR method on Meridian's `order_value` column across all customers together, and it flags **628 orders — 12.6% of the data**.

The Signal in the Flag Rate

Recall the calibration: on well-behaved data the $1.5 \times \text{IQR}$ fence should flag approximately **0.7%**. Getting 12.6% means one of two things: the data has a genuinely extreme distribution, or the assumption of a single population has failed.

In this case, it is the second. The file contains two fundamentally different customer types mixed together:

Segment	n	Median Order Value
Retail	4,414	₹1,471
Corporate	586	₹21,798

Those 628 "outliers" are corporate customers. Nothing is wrong with them. They are purchasing at the scale appropriate for their segment.

The Honest Caveat

The distribution of `order_value` as a single combined column does not obviously look like two populations. You could not have spotted the mixture from the shape of the histogram alone. It took a column already present in the file — `customer_segment` — and someone thinking to check it against the flag rate.

 When the fence flags far more than about 1% of a column, suspect your assumption about the population before suspecting the data. The fence is working correctly — the input was wrong.

Where Outlier Analysis Is Important

The method matters wherever the extremes either carry important information or corrupt everything downstream.



Data Quality

Catching typos, unit errors, sensor faults, and placeholder values before they corrupt every summary statistic, model, or dashboard built on top of them.



Fraud and Security

Unusual transactions, logins, and access patterns. Here the outlier is the *target*, not the nuisance. Removing it defeats the entire purpose.



Manufacturing and Quality Control

Units outside tolerance, drifting processes, defective batches. Statistical process control is built on the IQR method and its variants.



Operations

Severe delays, long outages, and failed deliveries — the events that customers actually notice and that generate complaints, churn, and rating drops.



Healthcare

Abnormal lab values, unexpected drug reactions, unusual patient outcomes. Outliers here are clinical alerts, not statistical inconveniences.



Finance and Risk

Extreme market moves, unusual trading patterns, claims far outside the normal range. Risk is fundamentally a question about the tail, not the centre.

IQR or Z-Score?

Both methods attempt to identify values far from the centre of a distribution. Only one of them is robust to the presence of the very values it is trying to detect.

Property	IQR Fence	Z-Score ($\pm 3\sigma$)
Built from	Quartiles	Mean and standard deviation
Affected by outliers?	No	Yes, badly
Works on skewed data?	Reasonably	No
Assumes normality?	No	Yes
Minimum sample size	~20	~30

The Z-Score Failure on Meridian Data

The order value of 99 (₹ thousands) in our worked example has a z-score of only **2.64**. It does not clear a 3-sigma filter. The value inflated the standard deviation used to judge it — raising the bar it then had to clear. The outlier defended itself by corrupting its own test statistic. The IQR fence flagged it without difficulty, because quartiles cannot be moved by a single extreme value in the tail.

When Z-Scores Are Appropriate

Z-scores are appropriate only when the column is approximately normally distributed — verified by a histogram or normality test, not assumed. On symmetric, well-behaved data with no extreme values, the two methods produce similar results. On real business data — income, order values, delivery times, claims — they typically differ substantially.

- ❑ Default to IQR. Use z-scores only after checking the data is roughly symmetric. If you have to ask which to use, the answer is IQR.

Where Outlier Analysis Should NOT Be Used

Knowing when not to run the method is as important as knowing how to run it. Applying it in the wrong context produces misleading flags and potentially harmful decisions.

→ **When the Outlier Is the Subject**

Fraud detection, rare disease research, catastrophe insurance, safety incident analysis. Removing outliers in these domains deletes the entire study. The method exists to protect the rest of the data — not to remove what you came to find.

→ **On Mixed Populations**

Meridian's order value flagged 12.6% across all customers combined. The fence described neither the retail nor the corporate segment accurately. Segment the data first; apply the method within each group.

→ **On Heavily Skewed Data Without Adjustment**

A long right tail is a genuine feature of income distributions, insurance claims, and order values — not a defect. Expect many flags on the raw column; expect most of them to be real. Consider a log transform before applying the method.

→ **On Small Samples**

Below roughly 20 observations, quartiles are unstable. A single value can move Q1 or Q3 substantially, and the resulting fence is not reliable. Report the raw values individually instead.

→ **On Categorical or Ordinal Data**

Quartiles require a meaningful numeric ordering and meaningful subtraction between values. Payment method, product category, and customer tier have neither. Do not apply IQR fences to these columns.

→ **Where Records Must Be Retained**

Regulatory, audit, clinical, and legal data cannot have observations quietly removed regardless of what the statistics indicate. Flag for review; do not delete.

The Blind Spot: Outliers in Combination

Everything covered so far checks one column at a time. Some outliers are only visible when you look across two columns simultaneously — and the IQR method on individual columns will miss them entirely.

The Meridian Example

Consider an order with:

- `units = 1` → perfectly ordinary by itself
- `order_value = ₹180,000` → within the normal range for corporate
- **Together** → a single unit costing ₹180,000, which no catalogue item is priced at

Neither column flags it independently. The combination is impossible. A ratio check on `order_value / units` finds it immediately.

Other Examples

- A patient whose height and weight are each individually normal but jointly imply an impossible BMI
- A transaction whose amount and location are each ordinary but have never been seen in that combination
- A delivery with a plausible time but an impossible distance-to-hub ratio

What to Do

Check ratios and derived columns. Apply the IQR method to engineered features — `value_per_unit`, `weight_per_item`, `time_per_task` — rather than only to the raw input columns. The ratio check catches most multivariate anomalies and requires only one additional line of code.

Formal multivariate methods also exist and are worth knowing:

- **Mahalanobis distance** — measures how far a point is from the centroid of the data in all dimensions simultaneously, accounting for correlations
- **Isolation forests** — a tree-based model that identifies points requiring fewer splits to isolate, which tend to be anomalies

 The ratio check catches most of what matters and is explainable to any stakeholder. Start there before reaching for more complex methods.

Never Do This

A practical checklist of the most common errors — each one documented in real analyses, often with significant consequences downstream.

Automated Deletion in a Pipeline

A job that flags and drops without human review will eventually delete your most valuable records. Flag and quarantine instead. Require a human decision before any row is removed.

Removing Outliers to Improve Model Performance

If a model only works once the difficult cases are removed, it does not work. You have trained it on the easy subset and it will fail exactly when you need it most.

Deleting First, Investigating Later

Once a record is removed, the observation and the reason it was unusual are usually both gone. Investigate in place. Delete only if the investigation concludes deletion is appropriate.

Running the Method Without Plotting First

A histogram or scatter plot takes thirty seconds and can reveal bimodal distributions, obvious data entry patterns, or clear segmentation that would invalidate the method before you run it.

Comparing Outlier Counts Across Groups

Mumbai and Delhi showed exactly why. A higher count can mean a tighter, better-controlled process — not more problems. Always compare the IQRs and fences that produced the counts.

Assuming a Flag Means an Error

Most flags are not errors. On real business data they are usually rare-but-genuine observations or evidence of a mixed population. The flag is a referral, not a verdict.

Checklist and Cheat Sheet

Three checkpoints, the code, and three numbers to carry out of this session.

Before Flagging

- Plot a histogram of the column
- Confirm $n \geq 20$
- Check for obvious bimodality or mixing
- Check the skewness — consider a log transform if skewness > 2

After Flagging

- Is each flagged value *possible*?
- Is it *plausible* given this business context?
- Would removing it *change the conclusion*?
- State your treatment decision in the methodology

Three Numbers to Remember

0.7%

Normal flag rate

What $1.5 \times \text{IQR}$ flags on well-behaved data. Far more means check your assumption.

2.64

Z-score of an obvious outlier

Why z-scores fail on skewed data — the outlier raised its own bar.

The Code

```
q1 = df['delivery_days'].quantile(0.25)
q3 = df['delivery_days'].quantile(0.75)
iqr = q3 - q1

# Use 3.0 for extreme outliers
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr

flagged = df[
    (df['delivery_days'] < lower) |
    (df['delivery_days'] > upper)
]

# Visualise by group
df.boxplot(
    column='delivery_days',
    by='fulfilment_hub'
)
```

- ☐ The default is keep. Removing data requires justification. Keeping it does not. When in doubt, report both the median and the IQR alongside any mean, and let the reader see the full picture.